

# Rare scenarios in germ cell tumors: A case series

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## Abstract

Germ cell tumors are native to the testis, retroperitoneum, and mediastinum. Here, we report our experience on rare scenarios in this entity. The first patient presented with a primary mediastinal germ cell tumor with 2° hemophagocytosis. This dysregulated immune milieu portends a dismal outcome irrespective of treatment. Gastrointestinal involvement in germ cell tumor accounts for only 5% of cases. The Second case with duodenal involvement and hematemesis is believed to be due to contiguous infiltration of the retroperitoneal node. Another case with bleeding ileal intussusception was seen after high-dose chemotherapy in a relapsed non-gestational choriocarcinoma. Bone metastases are exceptional in seminomas and we report such a case as well. Our next patient had thrombotic cardiac involvement which is scarce in the literature. Last patient in this series is an 81-year-old man with sertoli leydig cell tumor. This rare disease in elderly should be differentiated from other entities as the management for each differs.

## Keywords:

Germ cell tumor, hemophagocytosis, non gestational choriocarcinoma, sertoli leydig cell tumor

## Introduction

As per Globocan 2020 statistics, the global incidence of germ cell tumors (GCTs) is around 74,500 new cases and about one-third of them are diagnosed in Europe.<sup>[1]</sup> GCT forms the majority of testicular tumors in the adolescent and young adult men population. The incidence of testicular GCT is low in India. Nonetheless, due to their inherent chemosensitivity and curability, each rare patient deserves clinical mention. Mediastinal GCT with different poor risk biology has a dismal two-year survival. Here we report our experience with rare scenarios encountered in GCT at our tertiary cancer centre.

## Case 1: Primary Mediastinal GCT Presenting as Hemophagocytosis

A 22-year-old male was evaluated for fever, cough and breathlessness for one month duration. His

Eastern Cooperative Oncology Group performance status (ECOG-PS) was three and chest X-ray showed an anterior mediastinal mass [Figure 1a]. On Computed Tomography (CT) chest, the 8 × 7 × 5 cm anterior mediastinal mass was invading the aorta and superior vena cava. His Hb was 9.4 g/dl, total white blood cell count was 68,900/cmm and platelet count was 22,000/cmm. Total bilirubin was 9.6 mg/dl, SGOT and SGPT was 163 and 166 U/L, respectively, and lactate dehydrogenase (LDH) was 2576 U/L. The serum beta-human chorionic gonadotropin (β-HCG) was 53 mIU/ml and serum alfa fetoprotein (AFP) was 7200 ng/ml. His peripheral blood smear had a leukoerythroblastic blood picture, and bone marrow biopsy revealed histiocytes with features

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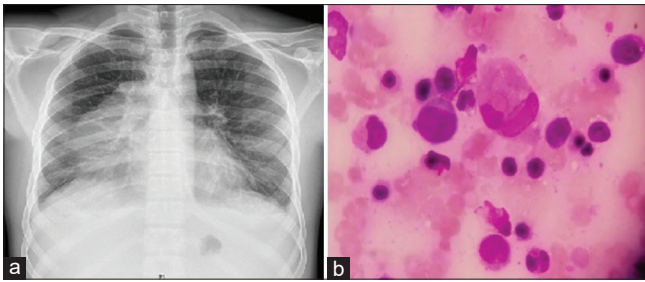
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**Figure 1: (a) Chest X-ray showing lobulated mediastinal mass; (b) Bone marrow aspirate (Geimsa stain  $\times 100$ ) – histiocyte showing hemophagocytosis (histiocyte with engulfed RBC and a neutrophil)**

of hemophagocytosis [Figure 1b]. Ultrasound scan of the scrotum was within normal limits. The diagnosis of primary mediastinal GCT presenting with hemophagocytosis was entertained. He was initiated on intravenous steroids, hepatic supports, chemotherapy (carboplatin with area under the curve (AUC) 2) along with platelet transfusion and supportive care. However, his condition deteriorated suddenly and he expired on the 9<sup>th</sup> day of presentation.

### **Case 2: Testicular GCT with Extensive Tumor Thrombus of Vena Cava with Right Atrial Extension**

A 25-year-old male, chronic smoker and alcoholic, was evaluated for right testicular swelling of six months duration. He presented with an ECOG-PS of three and dyspnea at rest. He had an enlarged right testis, bilateral pedal edema, left lower zone crepitations, and left cervical lymph nodes of  $2 \times 2$  cm. CT scan of chest/abdomen/pelvis showed right testicular mass, multiple lower cervical, supraclavicular, mediastinal, retroperitoneal and right iliac lymph nodes, multiple lung metastases and subcapsular liver metastasis. The serum LDH was 2618 U/L,  $\beta$ -HCG was 275 mIU/ml, and AFP was 6.6 ng/ml. Biopsy of the iliac node suggested metastatic GCT. After first cycle of etoposide, ifosfamide and cisplatin (VIP) multiagent chemotherapy, he defaulted and reported after ten months with poor general condition and features of superior vena cava obstruction. Chest CT showed bilateral lung parenchymal metastases, largest measuring  $5 \times 4.6$  cm, mediastinal lymph nodes encasing the great vessels, and extensive tumor thrombosis involving inferior vena cava (IVC), and superior vena cava extending to right atrium [Figure 2]. The  $\beta$ -HCG was 336 mIU/ml and AFP was 44.7 ng/ml. Due to poor general condition, he received chemotherapy with etoposide  $100 \text{ mg/m}^2$  plus carboplatin under AUC 2 and achieved clinical benefit.



**Figure 2: Computed Tomography (CT) chest showing large mediastinal deposit with tumor thrombus in the superior vena cava and bilateral pulmonary metastasis**

### **Case 3: Non-Gestational Choriocarcinoma with Ileal Involvement**

A 17-year-old boy with relapsed testicular non-seminomatous germ cell tumors (NSGCT) was referred to our department for high-dose chemotherapy (HDC) and stem cell rescue. His metastatic brain lesions were controlled with stereotactic radiosurgery. The s.  $\beta$ -HCG was 6761 mIU/ml, s. AFP was 1.8 ng/ml, and LDH was 374 U/L. The present CT showed multiple lung nodules, largest in the right middle lobe measuring  $37 \times 33$  mm, heterogenous mediastinal/hilar nodes, and right adrenal lesion of  $44 \times 37$  mm, which suggested progressive disease. The patient underwent HDC with carboplatin and etoposide (standard dose) for three days followed by autologous stem cell rescue. On the sixth day after chemo infusion, he developed severe vomiting along with abdominal pain. He had grade four neutropenia (TC: 200/cmm) and grade three thrombocytopenia (platelet count: 32000/cmm) at that time. Ultrasound imaging showed a 4.5-cm long intussusception in small bowel with preserved vascularity suggestive of ileo-ileal intussusception. The patient was initially managed conservatively with fluids, ryle's tube aspiration and antibiotics. He had hematopoietic recovery on 9<sup>th</sup> post chemo day, (TLC: 900/cmm, ANC: 690/cmm, platelet count: 60,000/cmm) and underwent exploratory laparotomy with resection (approximately 8 cm) of ileal intussusception and anastomosis. Intraoperatively, there was a pigmented ulcerated lesion at the lead point of the intussusception, which histopathologically revealed metastatic malignant GCT composed of choriocarcinoma component with extensive hemorrhages [Figures 3a and 3b]. The mucosa was ulcerated and the subserosal fat was

involved. Postoperative period was uneventful and his symptoms abated.

#### Case 4: Testicular NSGCT with Duodenal Perforation

A 57-year-old male, an ex-smoker, was evaluated for left testicular mass and left-sided abdominal pain of one month duration. He had undergone left high inguinal orchidectomy before presenting to us. At presentation, he was of ECOG-PS one and a vague mass was palpable in the left side of the abdomen. Serum AFP was 7773 ng/ml,  $\beta$ -HCG was 1.8 mIU/ml, and LDH was 990 U/L. CT thorax and abdomen showed a heterogeneously enhancing left paraaortic mass, measuring  $12.6 \times 10 \times 11.5$  cm, encasing abdominal aorta/renal vessels, and was abutting surrounding bowel loops. Histopathological examination of the orchidectomy specimen showed malignant mixed GCT with seminoma and yolk sac tumor elements. The final diagnosis of NSGCT intermediate risk was made. The nadir serum AFP attained after four cycles of VIP chemotherapy was 39 ng/ml. Post chemotherapy CT showed mesenteric deposits and a residual  $4.2 \times 3.7$  cm left paraaortic nodal mass, encasing renal vessels and lost fat planes with third part of duodenum. The s. AFP rose to 510 ng/ml within four weeks of the nadir and was started on TIP chemotherapy regimen. He had progressive disease after third TIP and received Gem-oxaliplatin further. The most recent CT showed a  $10 \times 9 \times 7.5$  cm paraaortic nodal mass infiltrating the third part of duodenum when the patient presented with anemia and massive hematemesis [Figure 4]. He was stabilized conservatively and opted for further care at local hospital due to COVID logistics.

#### Case 5: Testicular Seminoma with Bone Metastasis

A 48-year-old male with a history of undescended left testis was evaluated for painful abdominal pain of six months duration. His ECOG-PS was two and had an enlarged left supraclavicular node of  $2 \times 2$  cm. A vague epigastric mass and a  $3 \times 2$  cm firm mass in the left inguinal canal were palpable. Left testis was absent in the scrotal sac and the right testis was palpated normally. Serum LDH was 785 U/L, s. AFP was 1.6 ng/ml, and s.  $\beta$ -HCG was 35 mIU/ml. CT scan of thorax and abdomen showed soft tissue retroperitoneal mass lesion extending from T12 to L5 vertebra level encasing aorta/celiac vessels/superior mesenteric artery, left inguinal testis and lytic lesions involving L1–L3 vertebral body [Figure 5]. Bone scan showed increased uptake in L1–L3 vertebral bodies suggestive of metastases. Histopathological examination of the supraclavicular lymph node showed metastatic seminoma, which

were immunoreactive for Ckit, PLAP, Oct 3/4, and cytokeratin. With the diagnosis of intermediate risk seminoma, he received systemic chemotherapy with four cycles of VIP regimen with zoledronic acid. PET-CT after eight weeks showed FDG non-avid residual retroperitoneal nodes and sclerotic lesion in L1–L3 vertebral body. He underwent left high inguinal

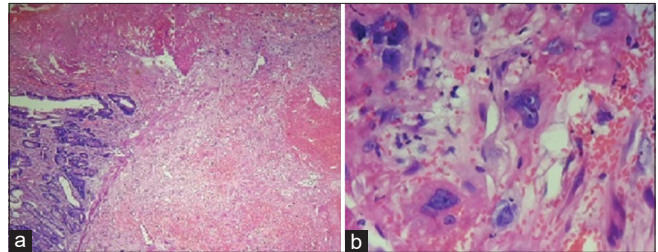


Figure 3: (a) Ileal tissue (H and E  $\times 100$ ) showing intestinal mucosa with underlying tumor; (b) Ileal tissue (H and E  $\times 400$ ) showing bizarre and pleomorphic tumor cells in a hemorrhagic background

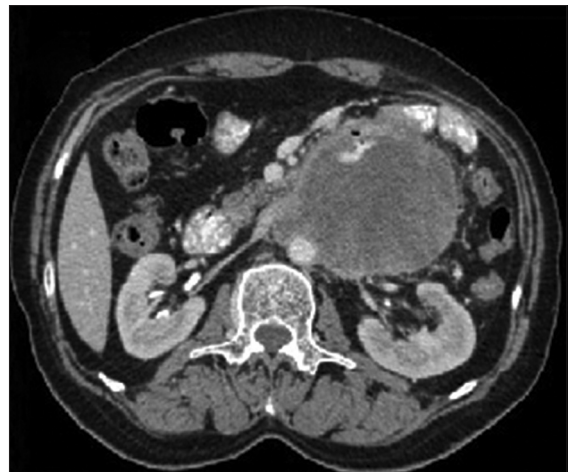


Figure 4: CT abdomen showing large left paraaortic nodal mass with necrosis, displacing and stretching the 4<sup>th</sup> part of duodenum anteriorly and communicating with the lower part of duodenum (oral contrast is seen entering into the necrotic nodal mass)

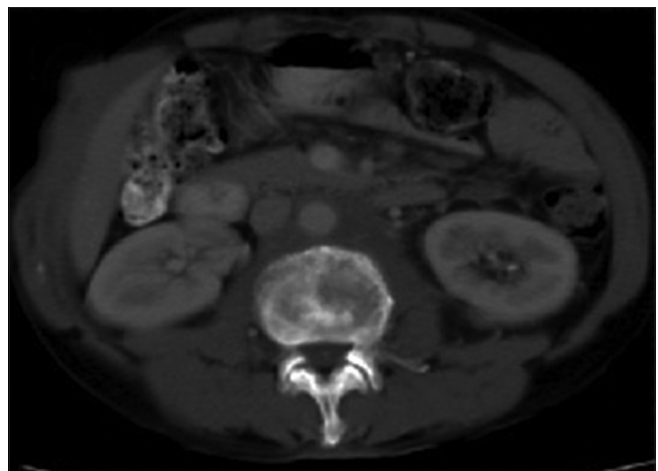


Figure 5: CT scan of pelvis showing mixed lytic sclerotic lesion involving the body of L4 vertebral body



orchidectomy and histopathological examination showed absent viable tumor cells. Patient is currently on follow up and is clinically well at three years.

#### Case 6: Leydig Cell Tumor in An 81-Year-Old Man

An 81-year-old man with no comorbidities was evaluated for left scrotal swelling. The clinical diagnosis was left hydrocele, and ultrasonogram of the scrotum showed a 1 × 1.2 cm echogenic lesion in the lower pole of the left testis. He underwent orchidectomy. Grossly, the testis measured 6 × 6 × 3 cm with a circumscribed mass of 1.3 cm diameter in the lower pole of testis. Histopathological examination was suggestive of Leydig cell tumor (LCT) testis, which were immunoreactive for calretinin and inhibin [Figure 6a–c]. There was no evidence of palpable or radiologic disease after one year of follow up.

#### Discussion

GCTs can primarily arise from gonads, retroperitoneum, and mediastinum. The mediastinal variety behaves differently as compared to the other two, in terms of treatment response and outcome. Male gonadal GCTs have been treated by time-tested protocols over the past three decades. Still, every rare scenario or complication needs to be highlighted, so that the practicing oncologists acquire an understanding of the same. In this paper, we have tried to gather the rare situations that we have encountered in our daily practice. In the following discussion, we have consolidated the discussion of each case with the relevant literature, which are updated to the best of our knowledge.

#### Primary mediastinal GCT presenting as hemophagocytosis

Mediastinal GCTs are known to be associated with hematological malignancies. A much similar

association of primary mediastinal GCT is secondary hemophagocytosis syndrome (HPS), which is due to continuous and unopposed stimulation of the patient's immune system. The abundant activated T/NK cells and macrophages elaborate huge amounts of pro-inflammatory cytokines like interferons, interleukins, tumor necrosis factor, etc., via unknown mechanisms, leading to HPS.<sup>[2]</sup> At the cellular level, erythrocytes, lymphocytes, and other hematologic precursors interact with Fc/C3b part of histiocytes and macrophages, whereby the former are eaten up and digested.<sup>[3,4]</sup> HPS is characterized by fever, coagulopathy, liver derangement, and pancytopenia.<sup>[2]</sup> The prognosis of such a patient is dismal, as in our case. The patient warrants emergency treatment, with steroids to combat immunodysregulation along with systemic chemotherapy. In majority of the patients, the course is stormy and outcome is poor. However, there are a few case reports where patients have responded to chemotherapy as well.<sup>[2]</sup> As per Taylor *et al.*<sup>[3]</sup> in a recent article, one among 11 patients with primary mediastinal GCT will develop hematological malignancy in the future, which suggests a clonal similarity between these two entities. In this study, majority shared p53/RAS alterations, indicating the possibility of a common precursor. This rare entity is a clinical challenge to the treating medical oncologist.

#### Testicular GCT with extensive tumor thrombus in major veins and right atrium

Thromboembolic events are common in malignancies; however, cardiac thrombosis is a rare event in testicular GCT. Only a limited number of such cases have been reported in the literature.<sup>[5–7]</sup> The largest autopsy series among testicular GCT at Memorial Sloan Kettering Cancer Center reports the frequency of IVC thrombosis as 11%, which is higher than the series from MD Anderson (3%).<sup>[8,9]</sup> Testicular GCT generally disseminates stepwise in a predictable lymphatic and hematogenous fashion. A step ahead, the right-sided tumors can spread through the right gonadal vein, which, by virtue of its direct communication with IVC, has an easy access to the same. It is also possible that the bulky lymphnodal conglomeration would have generated collateral-shunt mechanisms and directly eroded the intimal layers of the major vessels like IVC. In our case, both these factors contributed. The major complication of IVC thrombosis is a sudden massive embolism into vital organs including the heart. As per the autopsy series by Bredael *et al.*, 9% of patients with testicular malignancy had pulmonary embolism.<sup>[8]</sup> This implies that a high degree of clinical suspicion is necessary for these curable cancers, especially in the era of prophylactic anticoagulation. Positron emission tomography-computed tomography (PET-CT) can

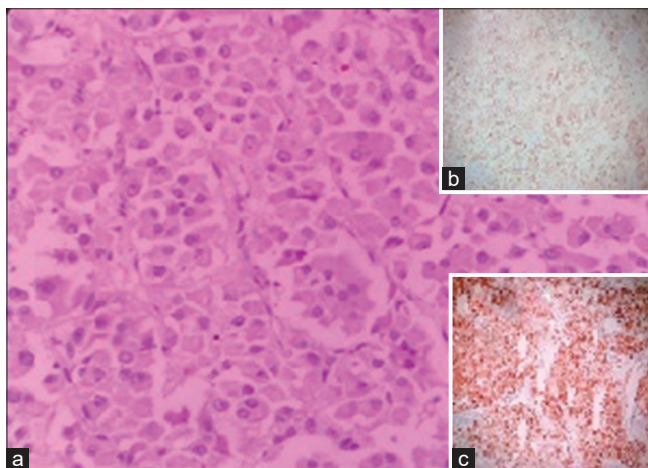


Figure 6: (a-c) Orchidectomy specimen (H and E × 40) Leydig cell tumor (a), IHC showing tumor cells positive for inhibin (b) and calretinin (c)

aid us in differentiating between a bland thrombus and a tumor thrombus. Tumor thrombus from GCT extending to the cardiac vessels or chambers is very rare, and its literature is scarce. Such patients preferably should undergo thrombectomy before initiating chemotherapy.<sup>[10]</sup> Our patient had physical fitness only for palliative chemotherapy. This case is highlighted for emphasizing the clinical suspicion of major vessel thrombosis, with or without cardiac chamber involvement, in GCT. Moreover, the thrombus is predicted to undergo chemoresolution, as they retain the chemosensitive nature of the primary disease.

#### **Testicular GCT with duodenal (1) and ileal (2) involvement**

Gastrointestinal tract (GI) dissemination from testicular GCT is seen in only 5% of cases, among which only 1.4% of cases involve the duodenum.<sup>[11]</sup> In the autopsy series by Johnson *et al.*, only 2 out of 78 patients had duodenal metastasis.<sup>[9]</sup> Chait *et al.*<sup>[12]</sup> studied 487 patients with testicular GCT in which only seven cases had duodenal involvement. Compared to seminomas, non-seminomas are more prone to GI dissemination. Among the GI sites, small bowel and duodenum are the commonest.<sup>[13]</sup> GI involvement in patients with metastatic GCT is mostly via tumor infiltration from affected retroperitoneal lymph nodes or via peritoneal seedling. The third part of the duodenum is a retroperitoneal structure and is at ease of spread/continued progressive involvement from the involved retroperitoneal lymph node, as proposed in our case. The symptoms may be abdominal pain, hematemesis/melena and consequent anemia. Another likely situation of duodenal bleeding in GCT is chemotherapy-induced necrosis. In such cases, in-patient care and meticulous support with anti-tumor lysis agents, intravenous fluids, and hemostatic agents are warranted. If a newly diagnosed patient presents with extensive metastatic disease (any organ with huge tumor load, poor general condition, bleeding tendencies, etc.), prophase chemo with etoposide and cisplatin (EP weekly for 2–3 cycles) before standard chemotherapy helps in preventing early treatment-related deaths.

Our second case with GI involvement was a young boy, who manifested with small bowel involvement during HDC and stem cell rescue. This is probably explained by the lysis of choriocarcinomatous cells under the effect of HDC. Of note, there is a subgroup of patients within poor risk NSGCT, called super-high-risk patients, who manifest with choriocarcinoma syndrome.<sup>[14]</sup> They are characterized by huge lung metastases, high  $\beta$ -HCG of more than 50,000 and are at risk of bleeding at the initiation of chemotherapy. However, this picture is not restricted

to pure choriocarcinoma alone, but can also be seen in other histologies like seminomas and embryonal carcinomas as well. A bleeding choriocarcinoma syndrome is generally anticipated at the treatment initiation or even in pretreatment state. It can also happen in the latter lines of treatments, as in our case. Choriocarcinoma cells have a high tendency to invade and outgrow the blood vessels, causing inherent necrosis. Chemotherapy administration in this situation carries high risk of bleeding which could be complicated by neutropenia. The risk of GI bleeding was not expected in our patient, as he was asymptomatic and not chemo-naïve. The patient had asymptomatic GI involvement at disease relapse which underwent lytic bleeding after HDC. He received aggressive supportive care for tumor lysis amidst grade 4 neutropenia. This case teaches us the possibility of organ bleeding in a patient with high  $\beta$ -HCG, irrespective of chemo-naïveness. An uncompromised supportive care is very essential in managing such patients.

#### **Testicular seminoma with bone metastasis**

Bone metastases in GCT are rare incidents, accounting for <10% at their primary diagnosis.<sup>[15]</sup> Their true incidence and response to stage adapted chemotherapy regimens are scarce. The Global Germ Cell Tumor Collaborative Group G3 Registration study evaluated 123 patients with bone metastases over a 30-year period. Histology-wise, non-seminomas comprised 77% and seminomas comprised 23% ( $n = 28$ ). According to them, the spine was the most common site, followed by pelvis, and they tend to be multi-sited. On multivariate analysis of seminoma vs non-seminoma, the hazard ratio was 0.33 ( $P = 0.011$ ), with a two-year progression-free survival and overall survival rates of 68% and 75% compared with 24% and 36%, respectively.<sup>[15]</sup> In another retrospective study published in 2013, 19 out of 2,550 patients had bone metastases, among which only three had seminomatous histology.<sup>[16]</sup> Our patient with testicular seminoma with spinal bone metastases received VIP chemotherapy along with bisphosphonates, as for intermediate-risk group. It seems that seminoma patients with bone metastases respond much alike the classically described intermediate-risk group seminoma; however, we need much more prospective database for conclusion to that effect. The local treatment to the affected bone is not guidelineed, but needs to be individualized (local radiotherapy or surgical stabilization), depending on other risk factors like spinal stability and pain.

#### **Leydig cell tumor in an 81-year-old man**

Testicular LCT is a rare entity. They account for less than 3% of testicular neoplasms and form the

commonest among interstitial testicular neoplasms.<sup>[17]</sup> The exact pathogenesis of LCT is still debated. One postulation is hypothalamic pituitary testis axis disruption leading to overactive luteinizing hormone and Leydig cells.<sup>[17]</sup> Cryptorchidism does not predispose to LCT. A bimodal age distribution can be seen in LCT, that is, prepubertal (5–10 years) and 30–60 years, although both extremes have been reported. The youngest patient reported is a 3.5-year-old boy and the eldest is a 91-year-old man, as per the published literature.<sup>[18]</sup> Only very few cases have been reported in >80-year-old age group and our 81-year-old patient adds to the present literature. These are one among the steroid secreting tumors, which have the potential to secrete androgens and estrogens. Prepubertal LCT is benign, but 10% of the adult cases can be malignant. The adult patients are usually asymptomatic, as in our case. As he was diagnosed post op, the hormone levels were not available. Tumor of >5 cm, necrosis, local/lymphovascular invasion, DNA aneuploidy, and high mitotic rate could suggest malignancy.<sup>[19]</sup> The tumor markers like BHCG, AFP, and LDH are not elevated in pure LCT. They are resistant to chemotherapy and radiotherapy, and surgical removal of the involved testis is the mainstay of treatment. One should be aware of the differential diagnoses of testicular mass in an elderly patient, such as primary lymphomas, spermatocytic seminomas, stromal tumors, plasmacytomas, and metastasis.<sup>[18]</sup> Benign entities like epidermoid cyst, Leydig cell hyperplasia, fibromas, and hemangiomas are also possible. Testicular lymphomas account for 85% of the testicular lesions above the sixth decade of life, and they need to be distinguished from LCT. Among lymphomas, diffuse large B cell lymphomas comprise 90%. Painless testiculomegaly  $\pm$  retroperitoneal nodal involvement, fever/drenching night sweats/weight loss (B symptoms), normal  $\beta$ -HCG, AFP along with elevated LDH favour testicular lymphomas. On immunohistochemistry (IHC), immunopositivity for LCA and B cell markers alongside bcl-2/myc rearrangements favor lymphomas; whereas, LCTs will be immunoreactive for inhibin, vimentin, melan-A, and calretinin, and negative for LCA and B cell markers. Spermatocytic seminomas are GCTs which are typical to the elderly. They neither metastasized nor have a germ cell neoplasia *in situ* component. Gain of chromosome 9 and immunoreactivity for ckit and NY-ESO-1 are the characteristic features of spermatocytic seminomas.<sup>[19]</sup>

To summarize, these individual cases add to the present literature on rare events in GCT. These

clinical possibilities are to be kept in mind while treating this cohort of patients with GCT.

#### Declaration of patient consent

The authors certify that they have obtained all appropriate patient consent forms. In the form the patient(s) has/have given his/her/their consent for his/her/their images and other clinical information to be reported in the journal. The patients understand that their names and initials will not be published and due efforts will be made to conceal their identity, but anonymity cannot be guaranteed.

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#### Conflicts of interest

There are no conflicts of interest.

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